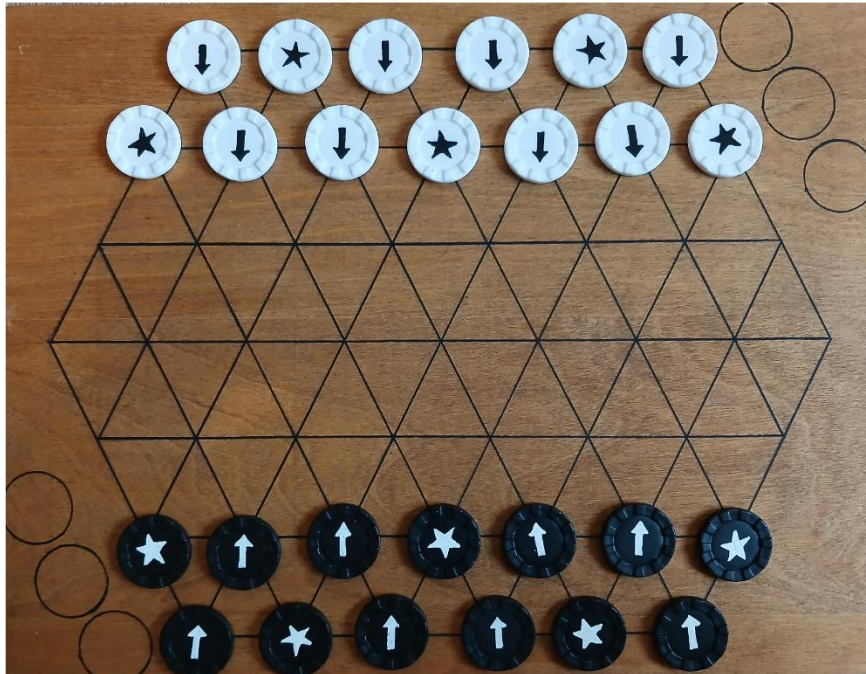


Jeu des Navettes (Shuttle Game)

2025, Xavier Bultel, <https://bultel.info/Navettes.html>

The game is played on the following board and setup:



arrow pieces are called “shuttles”, the other are called “stars”

On each turn, a player must perform the following two actions in order:

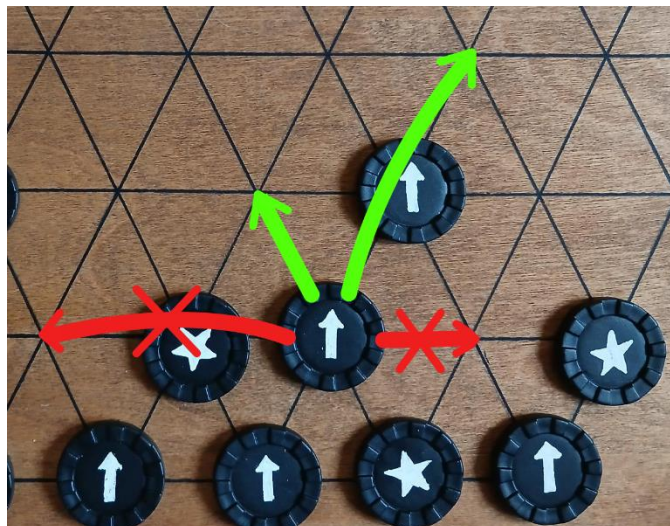
- 1) The player chooses one of their shuttles and moves it in one of the following two ways:
 - a. either move the shuttle from one intersection to an adjacent empty intersection in the direction indicated by its arrow (the arrow always points toward one of the two camps, never sideways; therefore at most two intersections are possible for this move);
 - b. or move the shuttle by “jumping” over another piece, as in Chinese checkers (the shuttle moves in a straight line over two intersections to land on an empty intersection, passing over an occupied one). After a jump, if possible, the player may choose whether or not to make an additional jump with the same shuttle. The first jump must be made in the direction indicated by the arrow; subsequent jumps may also be made sideways, but never in the opposite direction of the arrow. The number of successive jumps is unlimited, but it is forbidden to jump over the same piece more than once. At the start of a turn, if a player can jump an opponent’s piece with a shuttle, they must do so (this rule applies only to the first jump of the turn).

When a shuttle reaches one of the ends of the board (the first row of either player's camp), it is immediately flipped (the arrow then points toward the opposite camp). If this happens after a jump, the player may continue making successive jumps after flipping the shuttle.

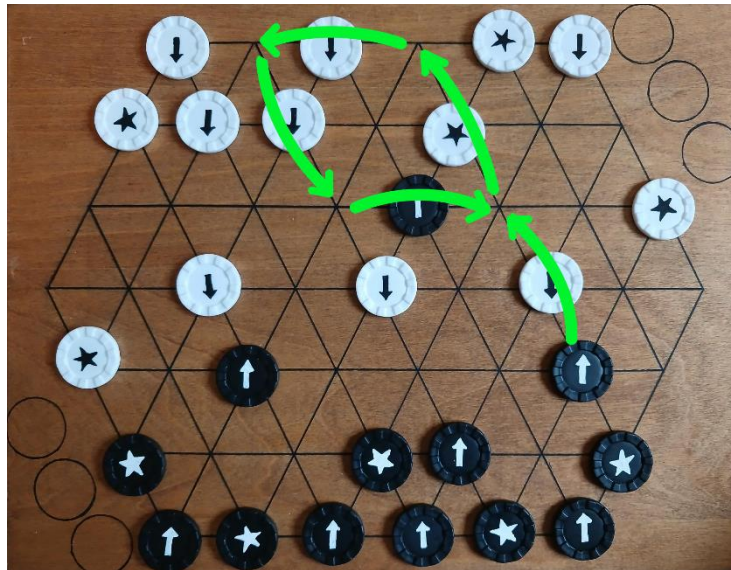
- 2) For this phase, the player has a number of moves equal to the number of opponent pieces jumped plus the number of times their shuttle flipped during the first phase of the turn. For each move, the player chooses one of their stars (the same star may be chosen multiple times) and performs one of the following actions:
 - a. either move the star from one intersection to an adjacent empty intersection toward the opponent's camp; or move the star by "jumping" over one or more pieces toward the opponent's camp (the star moves in a straight line to an empty intersection, passing over one or more intersections, all of which must be occupied). Note that a star may jump over several pieces in a single move, unlike shuttles. Be careful: if the player chooses to jump again with the same star, it costs an additional move.

A star may land "beyond" the last row of the opponent's camp. In that case, the star is removed from the game. Note: a star cannot leave the board via the left or right edges.

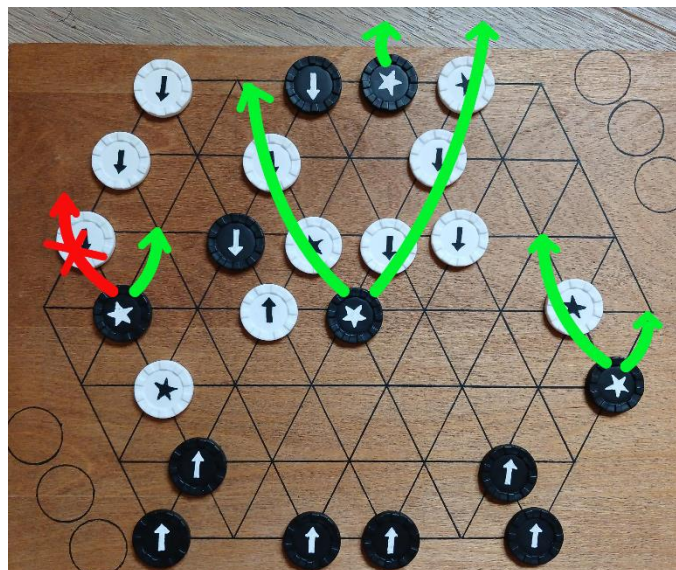
Goal. The game ends when all of one player's stars have left the board. That player wins the game.



In green, two legal shuttle moves (forward move and forward jump). In red, two illegal moves (sideways move and sideways jump as the first jump of the turn)



In this position, the player with the black pieces must jump an opponent's piece with one of their shuttles. The green arrows indicate a sequence of possible jumps for one of the black shuttles, allowing 5 star moves, because the black shuttle jumps over 4 white pieces and flips once (jumping over the black shuttle does not grant a move).



In this position, all possible moves of the black stars are shown with green arrows. In two cases, the star lands beyond the edge of the board and is therefore removed from the game.